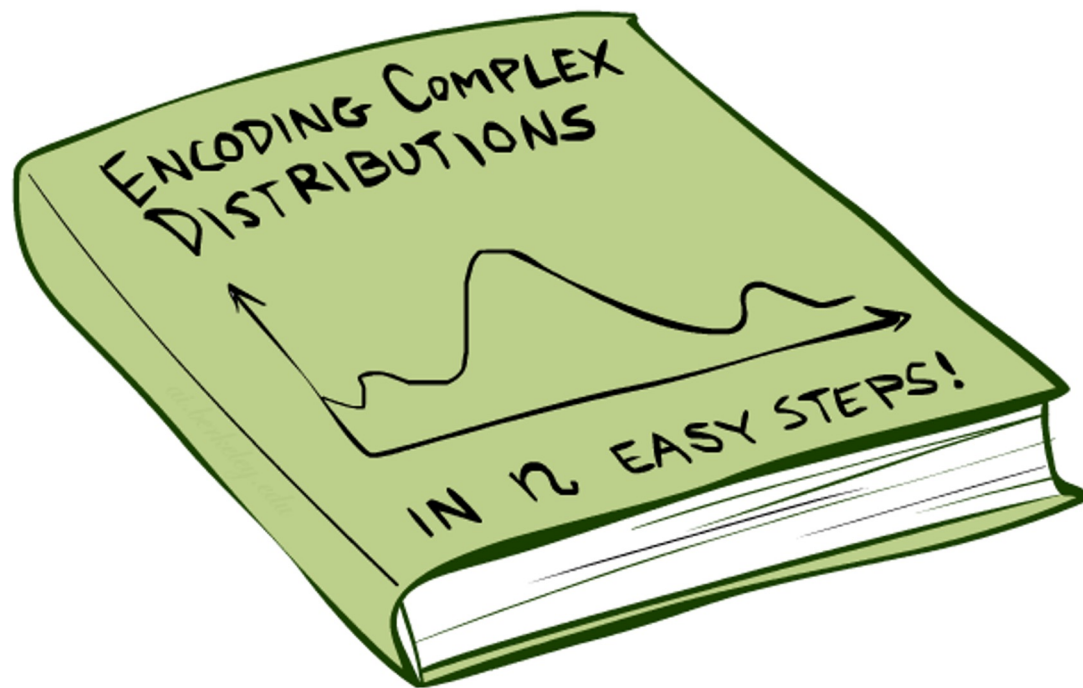


Bayesian Networks



Recap: Inference by Enumeration

General case:

- Evidence variables: $E_1 \dots E_k = e_1 \dots e_k$
 - Query variable: Q
 - Hidden variables: $H_1 \dots H_r$
- $$\left. \begin{array}{l} E_1 \dots E_k = e_1 \dots e_k \\ Q \\ H_1 \dots H_r \end{array} \right\} \begin{array}{l} X_1, X_2, \dots, X_n \\ \text{All variables} \end{array}$$

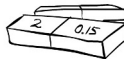
We want:

$$P(Q|e_1 \dots e_k)$$

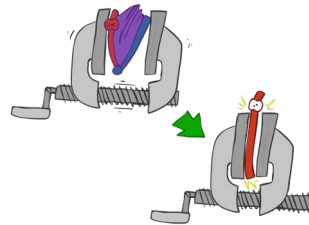
Step 1: Select the entries consistent with the evidence



x	P(x)
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01



Step 2: Sum out H to get joint of query and evidence



Step 3: Normalize

$$\times \frac{1}{Z}$$

$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} \underbrace{P(Q, h_1 \dots h_r, e_1 \dots e_k)}_{X_1, X_2, \dots, X_n}$$

$$Z = \sum_q P(Q, e_1 \dots e_k)$$

$$P(Q|e_1 \dots e_k) = \frac{1}{Z} P(Q, e_1 \dots e_k)$$

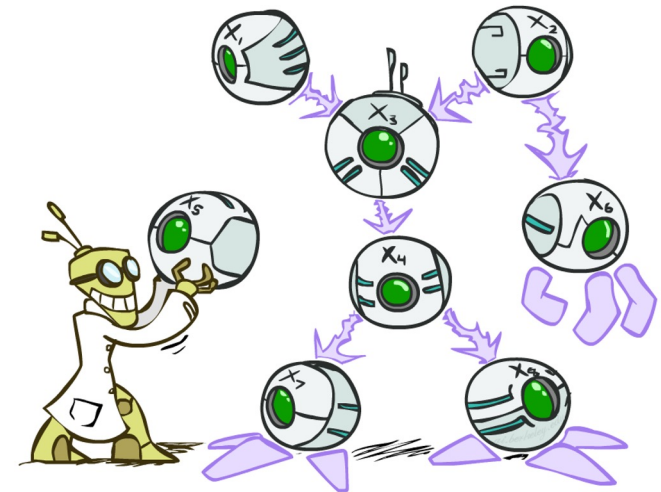
Bayesian Networks: Motivation

- Two problems with using full joint distribution tables as our probabilistic models for inference:
 - Unless there are only a few variables, the joint distribution is WAY too big to represent explicitly;
 - Hard to learn (estimate) anything empirically about more than a few variables at a time.



Bayesian Networks: The Big Picture

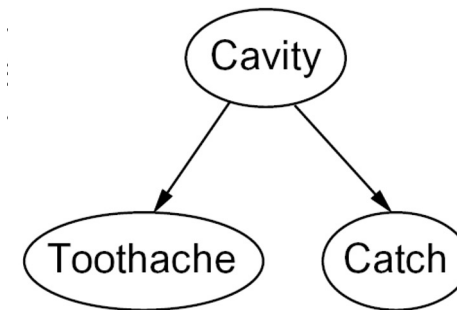
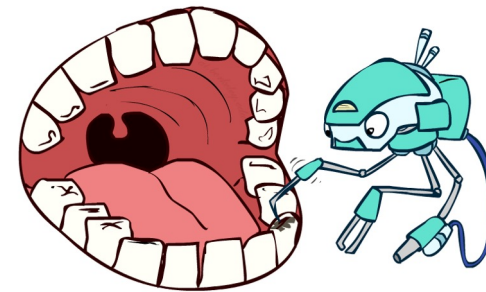
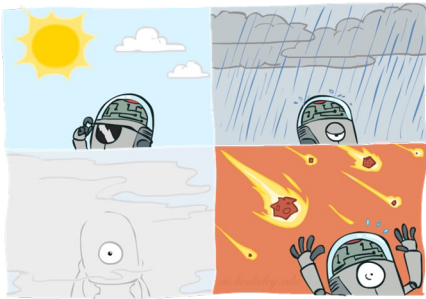
- **Bayesian Networks:** a technique for describing complex joint distributions (models) using simple, local distributions (**conditional probability tables**, or **CPTs**).
 - Using a directed graph to represent full joint probability distribution in a compact way.
 - More properly called **graphical models**.



Graphical Model Notation

- **Nodes**: corresponding to random variables (with domains).
 - Can be discrete or continuous variables.
 - Can be observed or unobserved (e.g., “hidden” variable).
- Directed links or **arrows**: connecting pairs of nodes.
 - Can describe how variables locally interact, or formally conditional independence relationship.
 - If there is an arrow from node X to node Y , X is said to be a **parent** of Y .
 - The graph has no directed cycles and hence is a directed acyclic graph (DAG).
 - Arrows mean that there may (but not necessarily) be a causal relationship between the two variables.

Bayes Net Examples

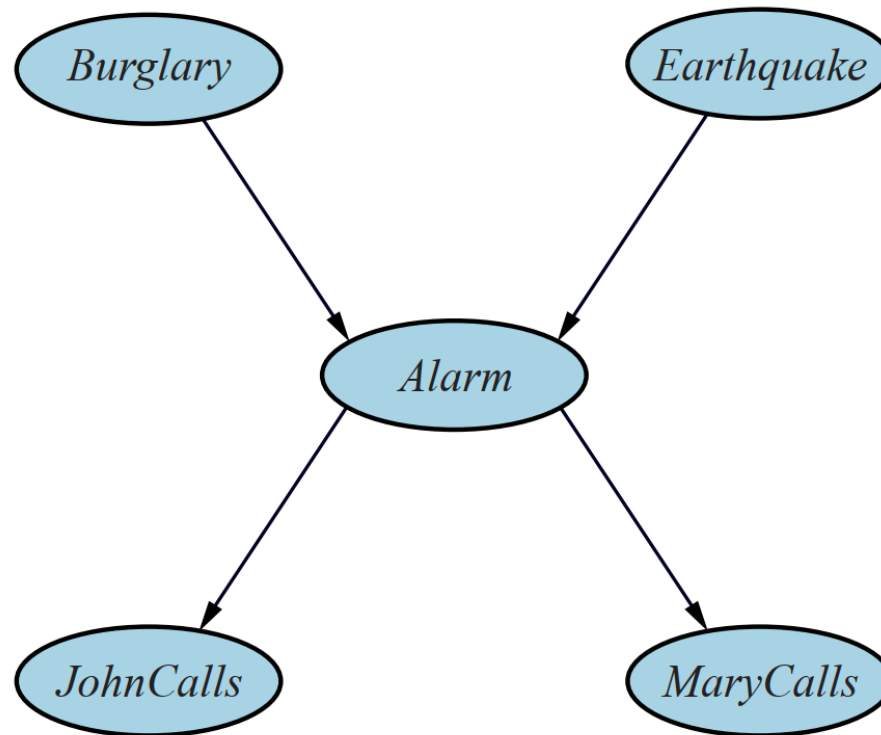


Naïve Bayes model can be represented by a simple Bayesian network.

Example: Alarm Network

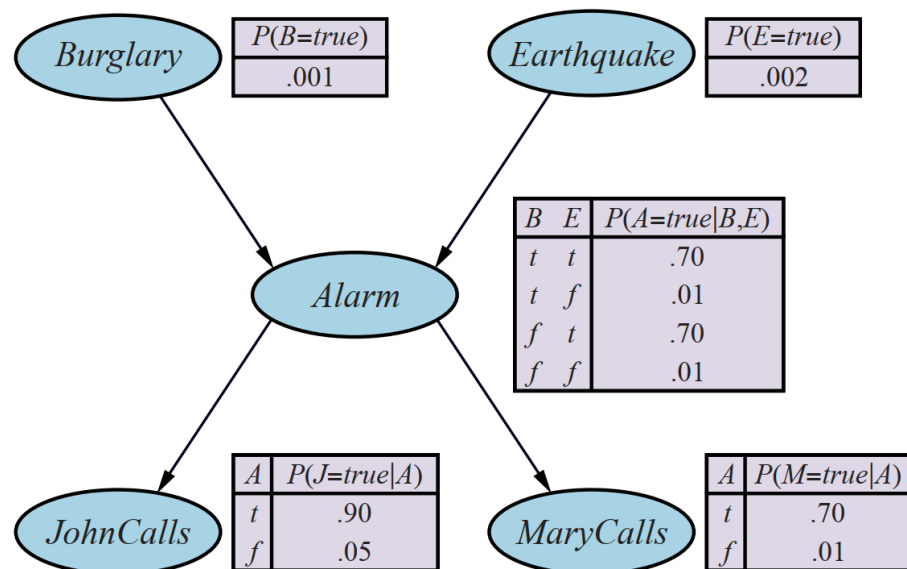
- Variables

- B: Burglary
- A: Alarm goes off
- M: Mary calls
- J: John calls
- E: Earthquake!



Conditional Probability Table (CPT)

- The local probability information attached to each node can take the form of **conditional probability table (CPT)**.
 - Each row in a CPT contains the conditional probability of each node value for a conditioning case.
 - CPT can only be used for discrete random variables as in this example.



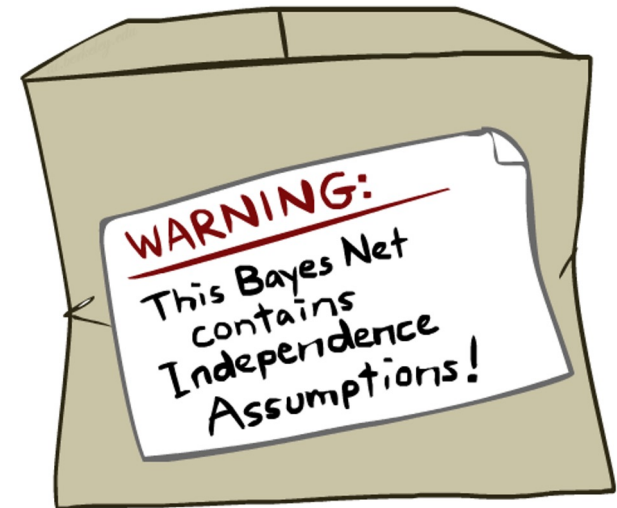
Are there **hidden** variables in this Bayes net?

Bayes Net Semantics

- The **syntax** of a Bayes net consists of a directed acyclic graph with some local probability information attached to each node.
- The **semantics** defines how the syntax corresponds to a joint distribution over the variables of the network.

$$P(x_i | x_1 \cdots x_{i-1}) = P(x_i | \text{parents}(X_i))$$

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

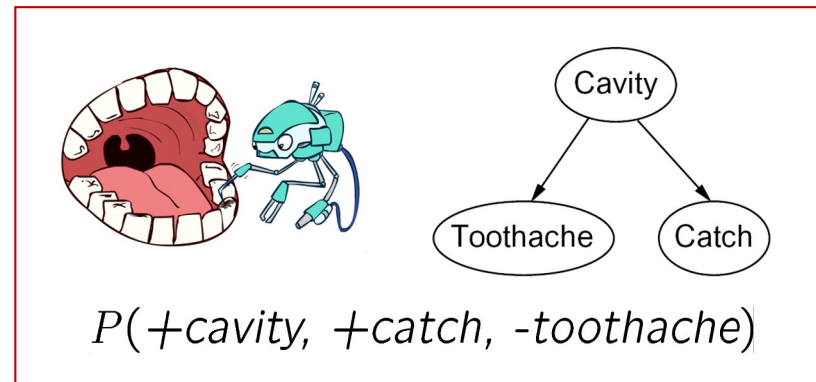


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- Recall that we can infer by using full joint distributions of all random variables. If a Bayes net is a representation of the joint distribution, then it can also be used to answer any query – which is a great news!

Constructing Bayes Net

- Assume there is an **ordering of all variables**, the **parents of node** X_i should contain all those nodes in $\{X_1, \dots, X_{i-1}\}$ that **directly influence** X_i .
- A Bayesian network accurately represents a domain **only if** each node is **conditionally independent** of its other preceding variables in the chosen ordering, **given its parents**.

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid \text{parents}(X_i))$$

vs.

$$\begin{aligned} P(x_1, \dots, x_n) &= P(x_n \mid x_{n-1}, \dots, x_1) P(x_{n-1} \mid x_{n-2}, \dots, x_1) \cdots P(x_2 \mid x_1) P(x_1) \\ &= \prod_{i=1}^n P(x_i \mid x_{i-1}, \dots, x_1). \end{aligned}$$

Constructing Bayes Net

1. Determine the set of variables that are required to model the domain.
2. Order all variables as $\{X_1, \dots, X_n\}$
 - Any order of the variables will work, but the resulting network will be more compact if the variables are ordered such that causes precede effects.
3. For $i = 1$ to n do:
 - Choose a minimal set of parents for X_i from $\{X_1, \dots, X_{i-1}\}$ to satisfy

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

- For each parent insert a link from the parent to X_i
- Write down the conditional probability table (CPT) for $P(x_i | \text{parents}(X_i))$

Example: Coin Flips

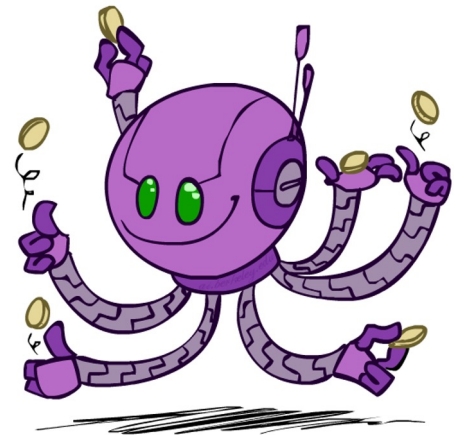
- N independent coin flips

X_1

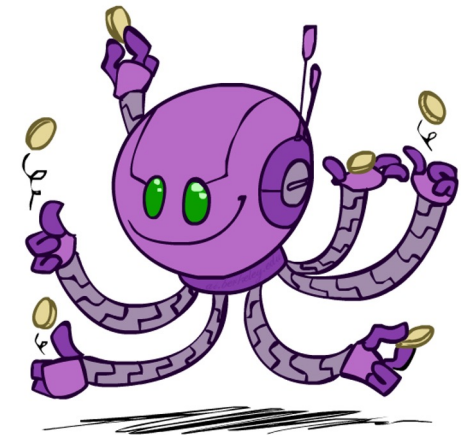
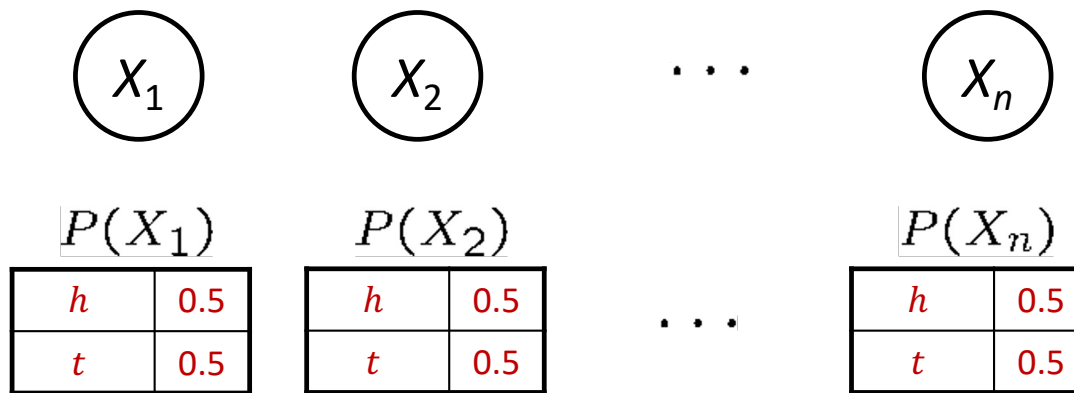
X_2

...

X_n



Example: Coin Flips

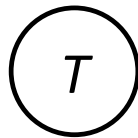
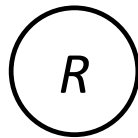


$$P(h, h, t, h) = P(h)P(h)P(t)P(h)$$

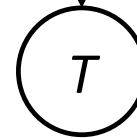
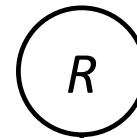
Only distributions whose variables are absolutely independent can be represented by a Bayes net with no edge links.

Example: Traffic

- Variables:
 - R : It rains
 - T : There is traffic



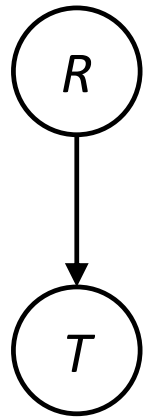
Model 1: independence



Model 2: rain may cause traffic



Example: Traffic

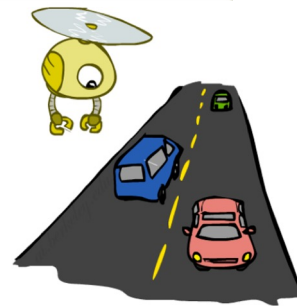

$$P(R)$$

$+r$	$1/4$
$-r$	$3/4$

$$P(T|R)$$

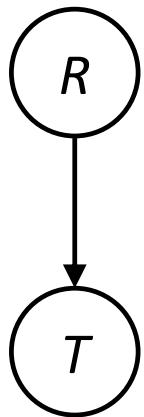
$+r$	$+t$	$3/4$
$+r$	$-t$	$1/4$
$-r$	$+t$	$1/2$
$-r$	$-t$	$1/2$

$$P(+r, -t) = P(+r)P(-t|+r) = 1/4 * 1/4$$



Example: Traffic

- Causal direction



$$P(R)$$

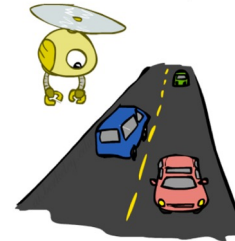
$+r$	$1/4$
$-r$	$3/4$

$$P(T|R)$$

$+r$	$+t$	$3/4$
	$-t$	$1/4$
$-r$	$+t$	$1/2$
	$-t$	$1/2$

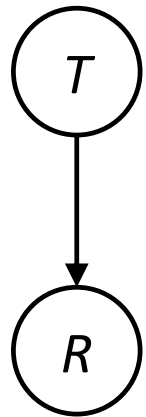
$$P(T, R)$$

$+r$	$+t$	$3/16$
$+r$	$-t$	$1/16$
$-r$	$+t$	$6/16$
$-r$	$-t$	$6/16$



Example: Reverse Traffic

- Reverse causality?



$$P(T)$$

$+t$	$9/16$
$-t$	$7/16$

$$P(R|T)$$

$+t$	$+r$	$1/3$
	$-r$	$2/3$
$-t$	$+r$	$1/7$
	$-r$	$6/7$



$$P(T, R)$$

$+r$	$+t$	$3/16$
$+r$	$-t$	$1/16$
$-r$	$+t$	$6/16$
$-r$	$-t$	$6/16$

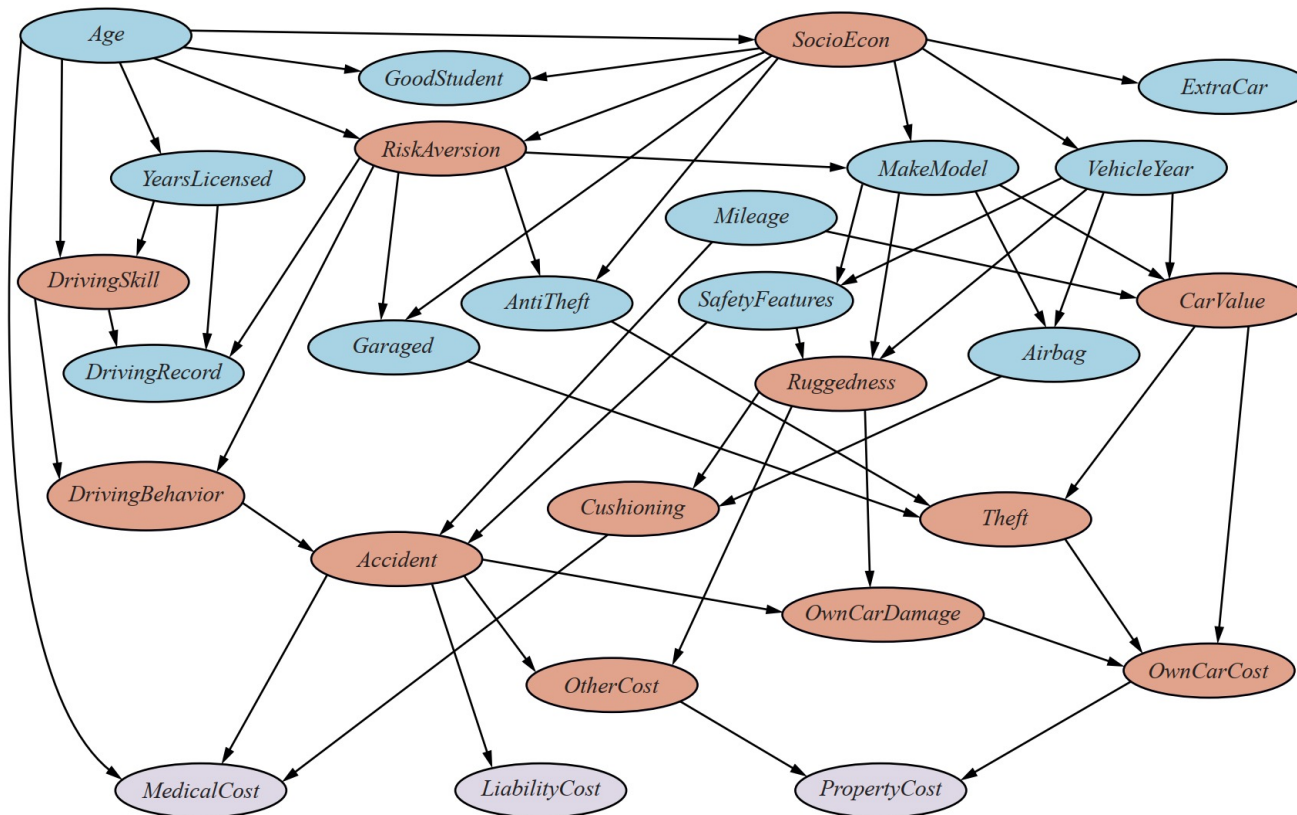
Causality in Bayes Net

- Bayesian networks do not necessarily encode causality.
 - The arrows may reflect statistical correlation, not causal influence.
 - In some domains, especially when variables are missing, a true causal network may not exist or be hard to define.
- What do the arrows really mean in Bayes nets?
 - The topology of a Bayes net primarily encodes conditional independence.
 - The structure may happen to align with causal relationships, but this is not guaranteed.

Causality in Bayes Net

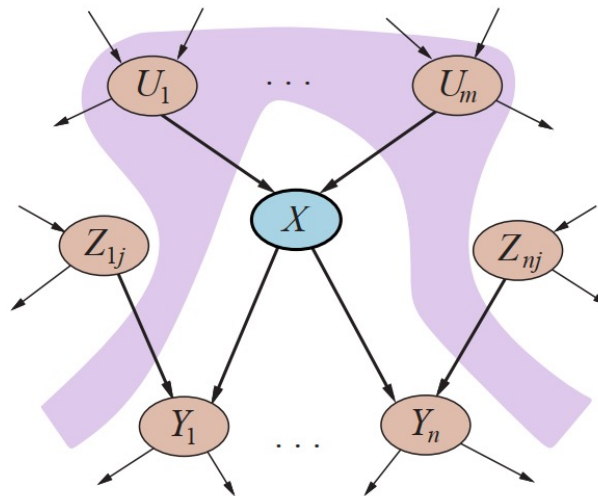
- When Bayes Nets Reflect True Causality,
 - The network is often simpler (fewer parents per node);
 - Easier to understand and interpret;
 - Easier to elicit from domain experts, especially in fields like medicine or engineering.

Example: Car Insurance



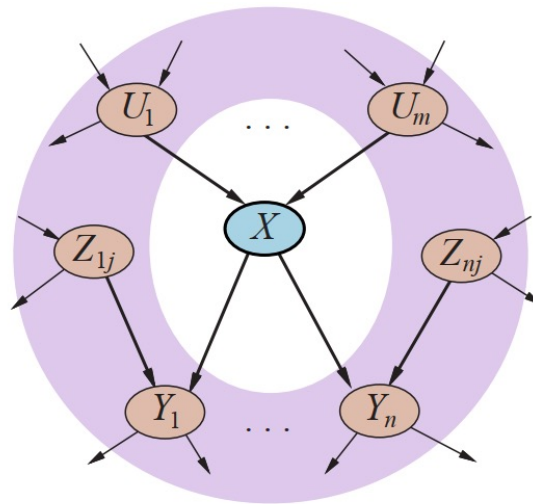
Conditional Independence in Bayes Net

- Each variable is conditionally independent of **its other predecessors** in the ordered variables, given its parents.
- Each variable is conditionally independent of **its non-descendants**, given its parents.



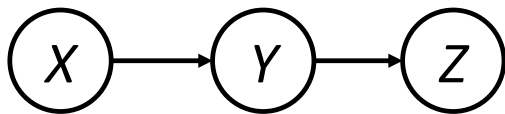
Conditional Independence in Bayes Net

- Each variable is conditionally independent of all other nodes in the network, given its **Markov blanket**.
 - **Markov blanket** refers to the parents, children, and children's parents.



Example: Causal Chain

- **Question:** Are X and Z guaranteed to be independent?
 - Answer: No (Example: low **pressure** causes **rain**, which further causes **traffic**)
 - X can influence Z , Z can influence X (via Y).



$$P(x, y, z) = P(x)P(y|x)P(z|y)$$



X: Low pressure

Y: Rain

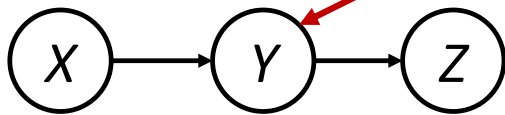
Z: Traffic

Example: Causal Chain

- **Question:** Is X guaranteed to be independent of Z , given Y ?

- Answer: Yes.

Evidence along the chain “blocks” the influence.



$$P(z|x, y) = \frac{P(x, y, z)}{P(x, y)}$$
$$= \frac{P(x)P(y|x)P(z|y)}{P(x)P(y|x)} = P(z|y)$$



X: Low pressure

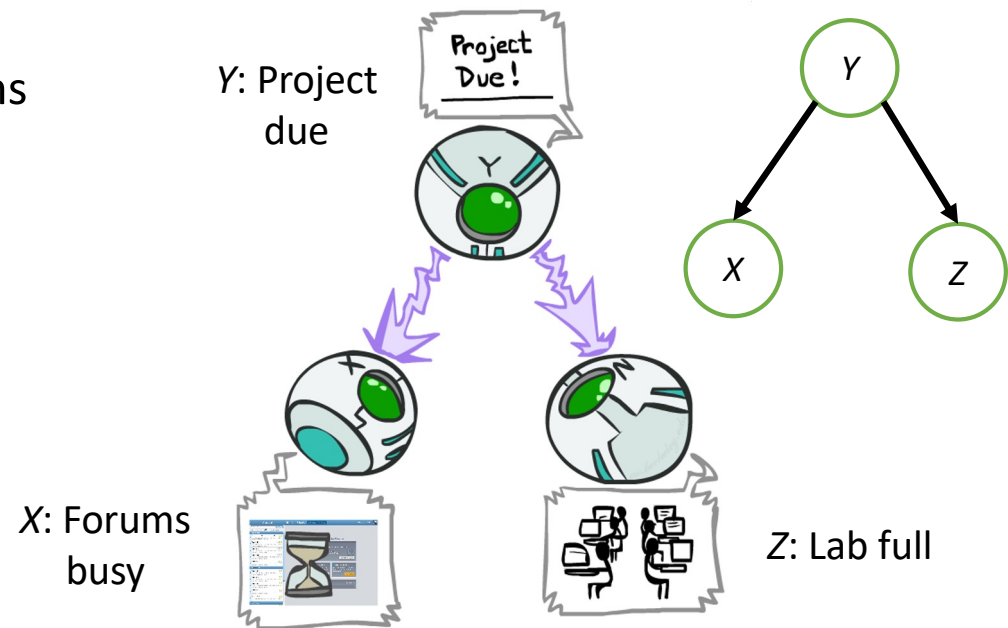
Y: Rain

Z: Traffic

Example: Common Cause

- **Question:** Are X and Z guaranteed to be independent?

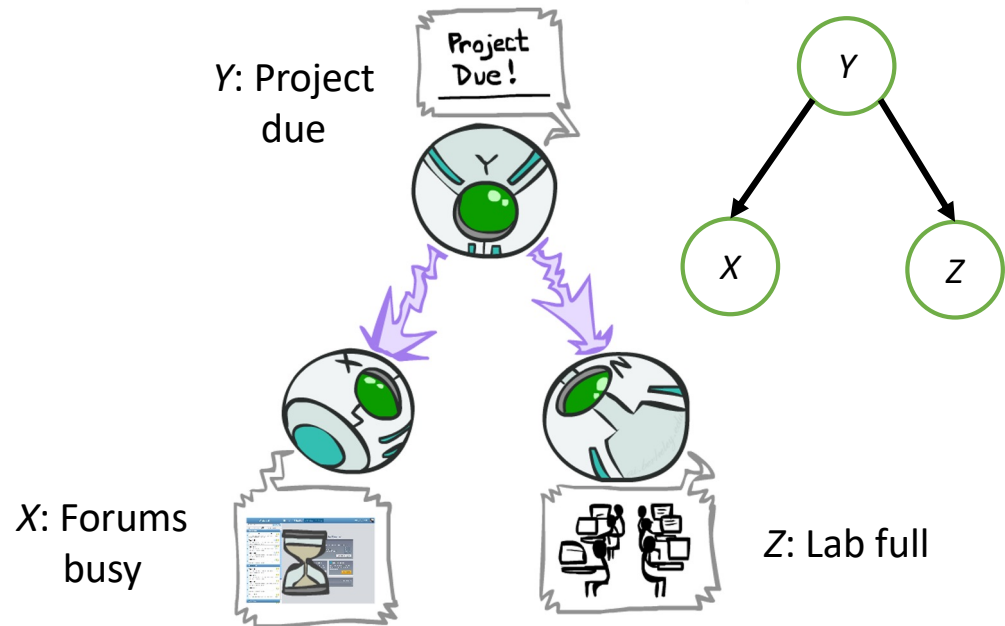
- Answer: No.
- E.g.: Project due causes both forums busy and lab full.



Example: Common Cause

- **Question:** Are X and Z guaranteed to be independent, given Y ?
- Answer: Yes.

$$\begin{aligned} P(z|x, y) &= \frac{P(x, y, z)}{P(x, y)} \\ &= \frac{P(y)P(x|y)P(z|y)}{P(y)P(x|y)} \\ &= P(z|y) \end{aligned}$$



Observing the cause blocks influence between effects.

Example: Common Effect

- **Question:** Are X and Y guaranteed to be independent?

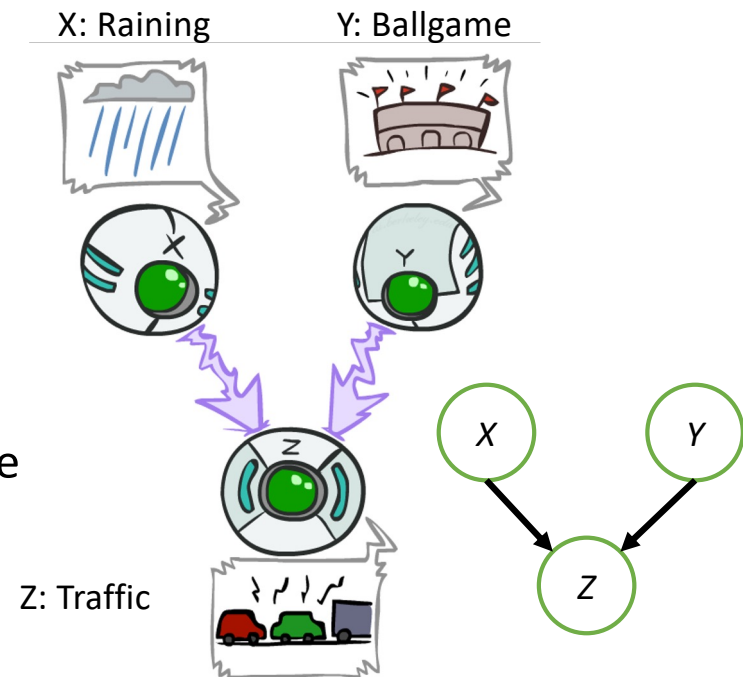
- Answer: Yes.
- E.g.: the ballgame and the rain cause traffic, but they are not correlated.

$$P(x, y) = \sum_z P(x, y, z)$$

- **Question:** Are X and Y independent given Z ?

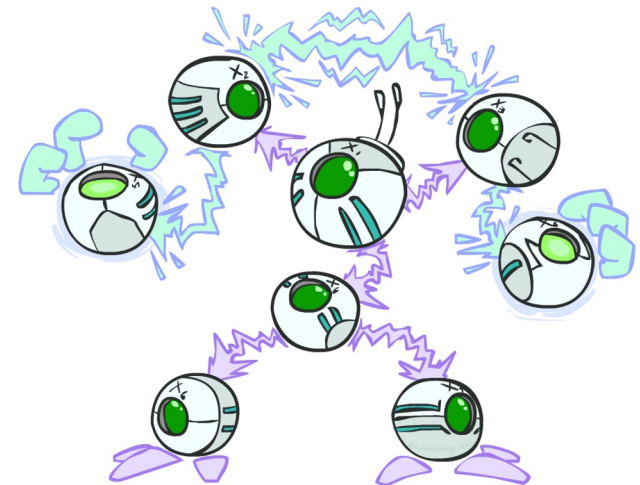
- Answer: No.
- E.g.: seeing traffic puts the rain and the ballgame in competition as explanation.

Observing an effect activates influence between possible causes.



The General Case

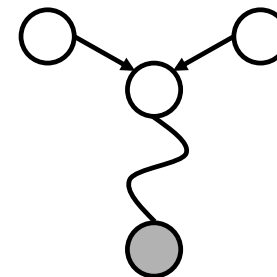
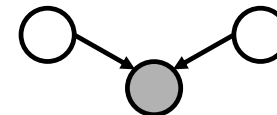
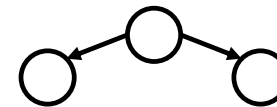
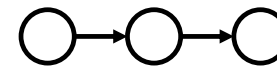
- General question:
In a given Bayes net, are two variables independent (given evidence)?
- Solution: analyze the graph.
- Any complex example can be broken into repetitions of the three canonical cases.
 - Causal chain;
 - Common cause;
 - Common effect.



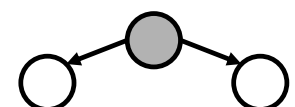
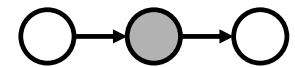
Active / Inactive Paths

- Question: Are X and Y conditionally independent given evidence $\{Z\}$?
 - Yes, if X and Y “d-separated” by Z .
 - Consider all (undirected) paths from X to Y .
 - No active paths = independence!
 - A path is active if each triple is active:
 - Causal chain $X \rightarrow Y \rightarrow Z$ where Y is unobserved.
 - Common cause $X \leftarrow Y \rightarrow Z$ where Y is unobserved.
 - Common effect $X \rightarrow Y \leftarrow Z$ where Y (or one of its descendants) is observed.

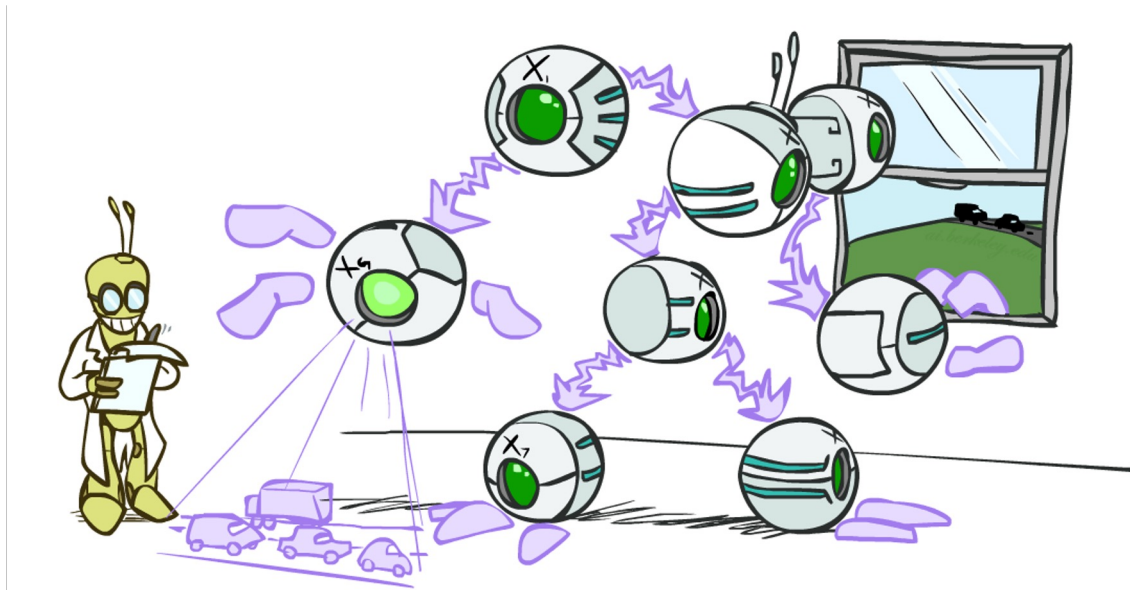
Active Triples



Inactive Triples



Inference with Bayes Net



Inference

- **Inference** is the process of computing the probability distribution of one or more variables, given observed evidence about other variables in the network.
 - It answers queries like “What is the probability of X given Y , or $P(X|Y)$?”
- Why inference is needed?
 - Reasoning under uncertainty: Diagnose diseases from symptoms; Detect faults in engineering systems; Assess risks in financial decision-making.
 - Predicting future events: Forecast traffic conditions based on weather; Estimate recovery outcomes based on treatments.
 - Identifying probable causes and initials: Trace the root cause of machine failure using sensor data; Understand underlying factors behind observed effects.

Example: Traffic Domain

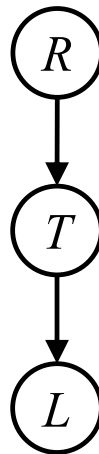
- Random Variables

- R: Raining
- T: Traffic
- L: Late for class!

$$P(L) = ?$$

$$= \sum_{r,t} P(r, t, L)$$

$$= \sum_{r,t} P(r)P(t|r)P(L|t)$$



$$P(R)$$

+r	0.1
-r	0.9

$$P(T|R)$$

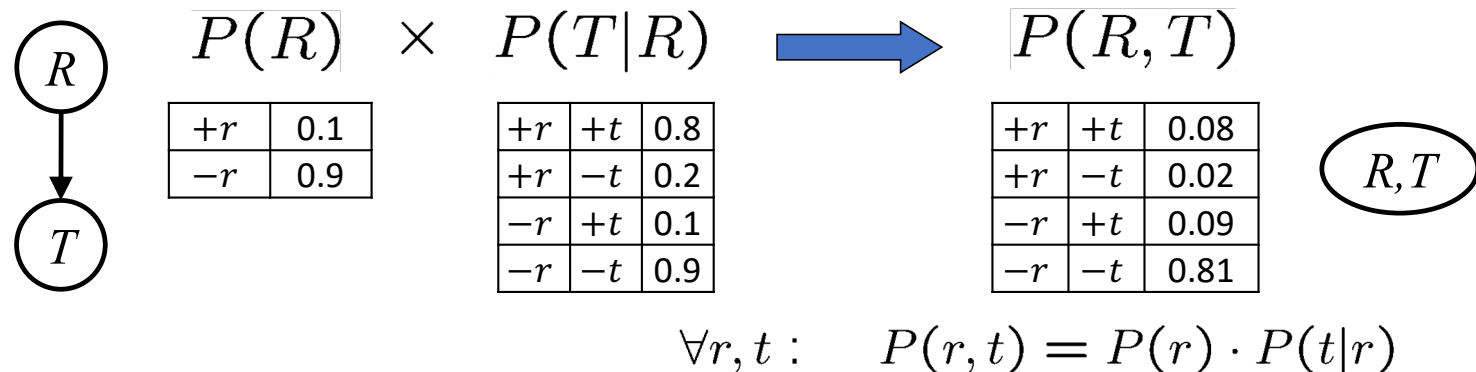
+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

$$P(L|T)$$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

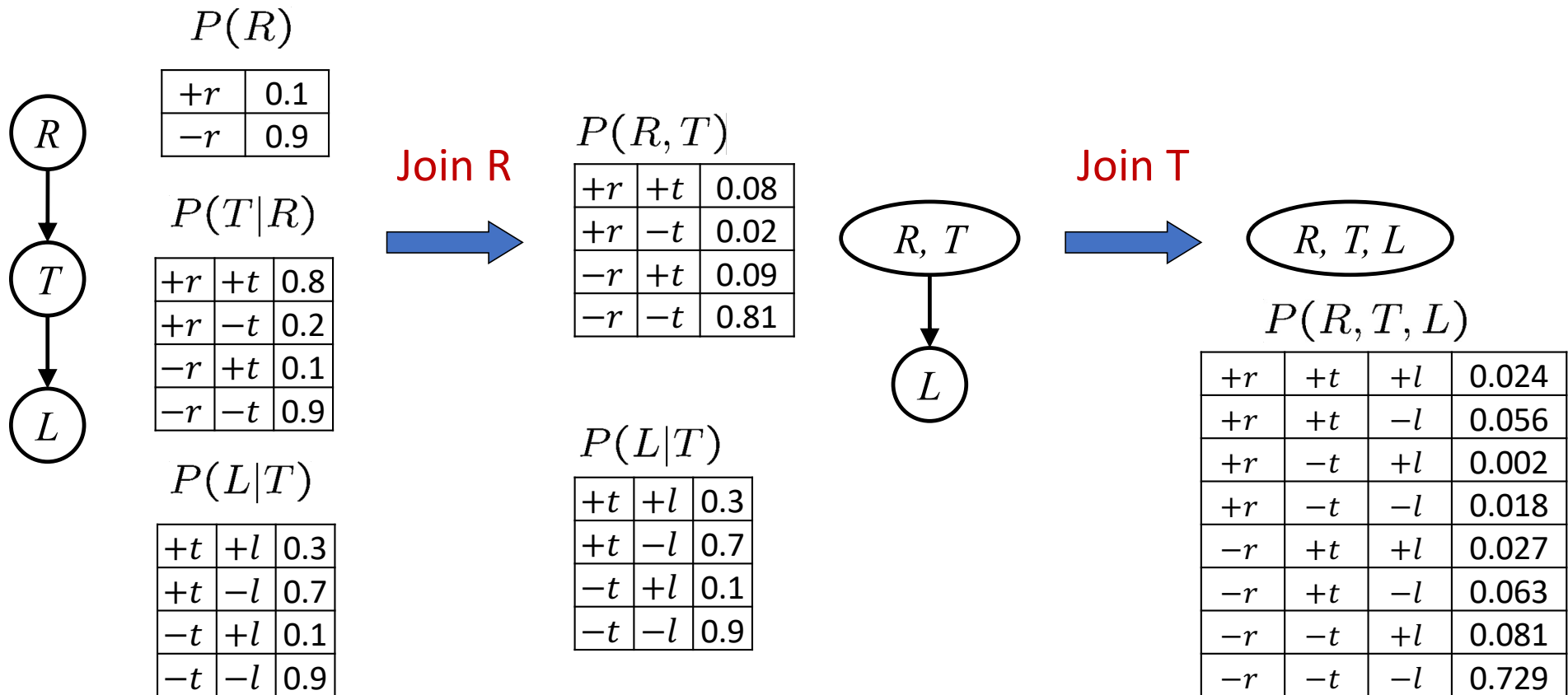
Operation 1: Join Factors

- First basic operation: **joining factors**
- Combining factors:
 - Get all factors over the joining variable;
 - Build a new factor over the union of the variables involved.
- Example: Join on R



- Computation for each entry: pointwise products

Example: Multiple Joins

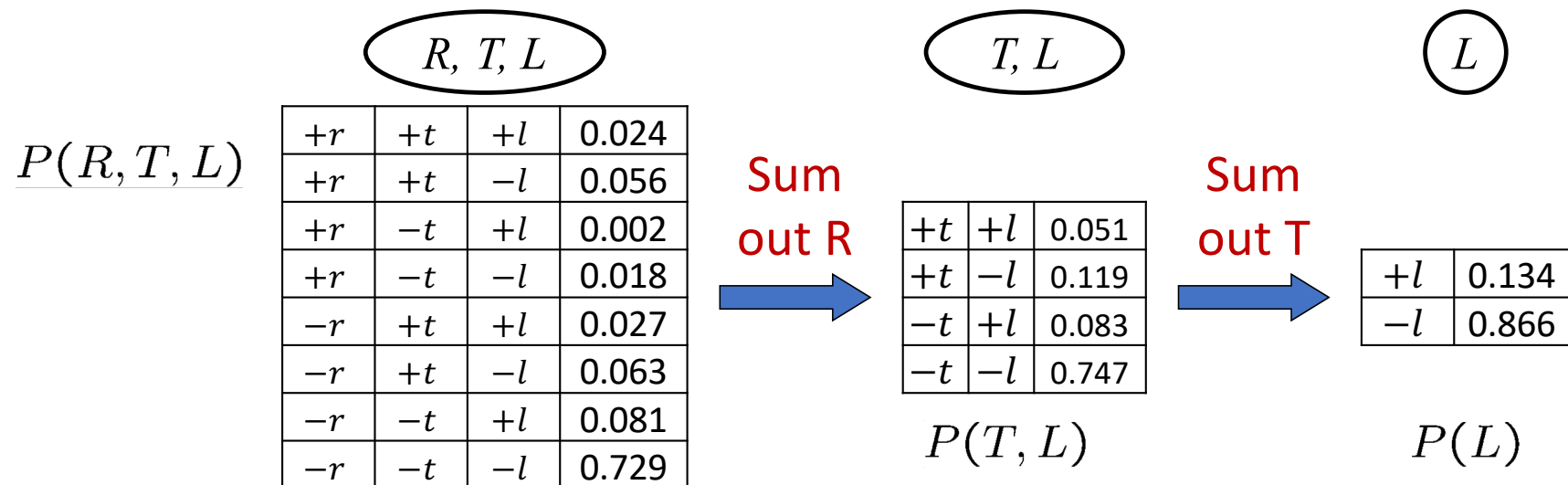


Operation 2: Eliminate

- Second basic operation: **marginalization**
- Take a factor and sum out a variable:
 - Shrink a factor to a smaller one;
 - A **projection** operation.
- Example:

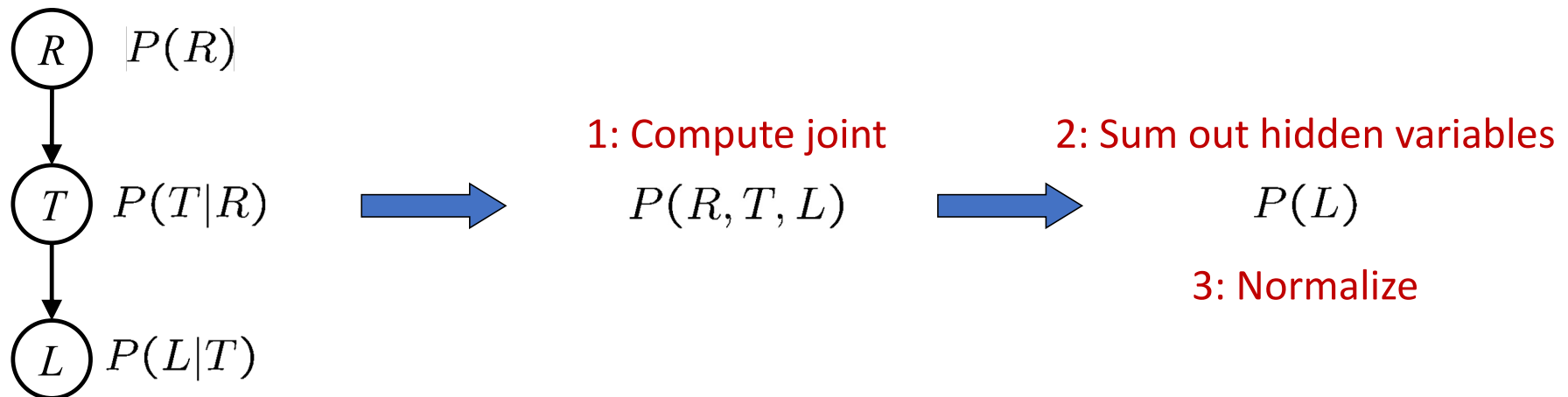
$P(R, T)$			Sum out R 	$P(T)$	
+r	+t	0.08		+t	0.17
+r	-t	0.02		-t	0.83
-r	+t	0.09			
-r	-t	0.81			

Multiple Elimination



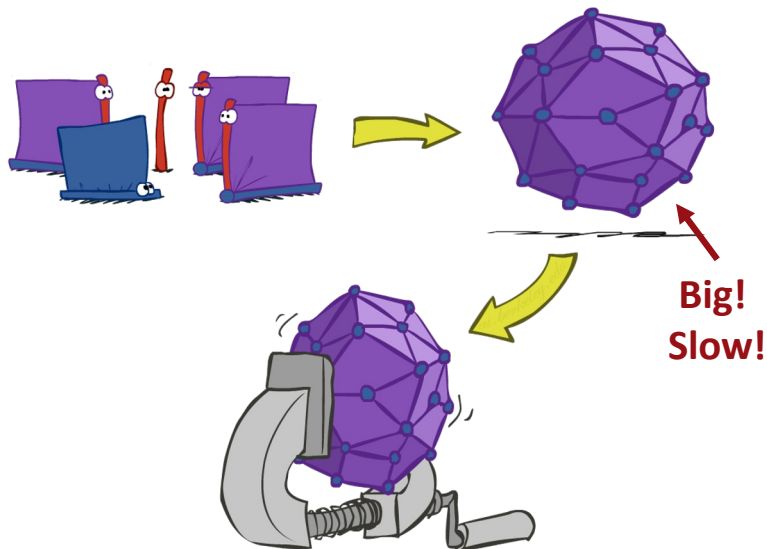
Multiple Join, Multiple Eliminate

- The procedure of multiple-join-multiple-eliminate is equivalent to inference by enumeration.

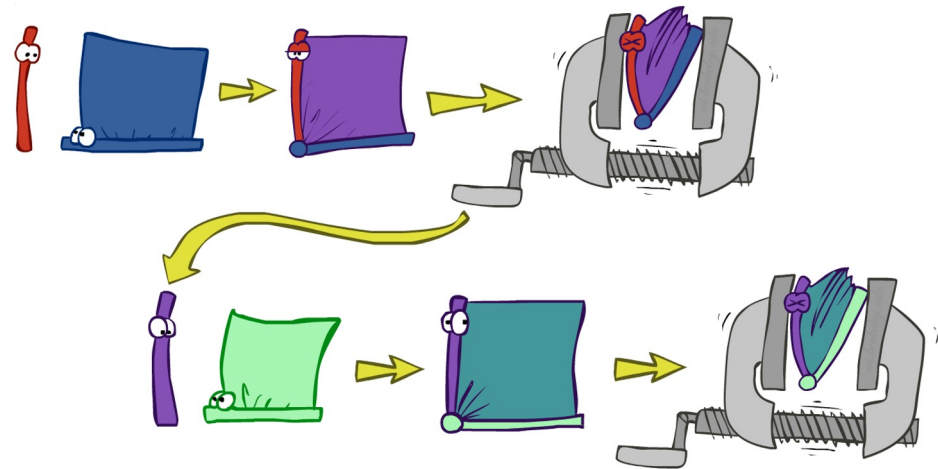


Variable Elimination

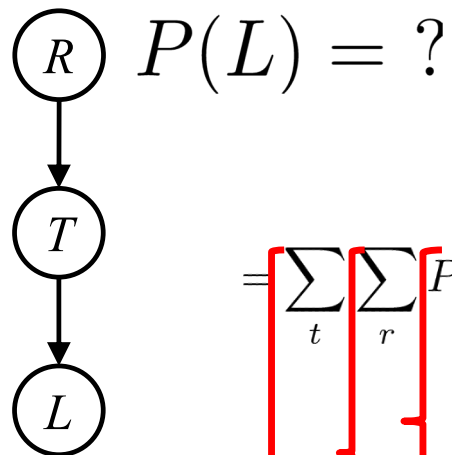
- Why is inference by enumeration slow?
 - Join up the whole joint distribution before you summing out the hidden variables.



- Idea: **interleave joining and marginalizing!**
 - Called "Variable Elimination".



Example: Traffic Domain



$$= \sum_t \left[\sum_r \left[P(L|t) P(r) P(t|r) \right] \right]$$

Eliminate t

Join on t

Eliminate r

Join on r

Inference by Enumeration

$$= \sum_t \left[P(L|t) \left[\sum_r P(r) P(t|r) \right] \right]$$

Eliminate t

Join on t

Eliminate r

Join on r

Variable Elimination

The computational and space complexity of variable elimination is determined by the largest factor (e.g., check $P(R,T,L)$).



$P(R)$

T	L
+r	0.1
-r	0.9

$P(T|R)$

+r	+t	0.8
+r	-t	0.2
-r	+t	0.1
-r	-t	0.9

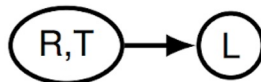
$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Join R

$P(R,T)$

+r	+t	0.08
+r	-t	0.02
-r	+t	0.09
-r	-t	0.81

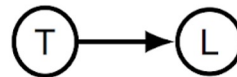


$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Sum out R

+t	0.17
-t	0.83



$P(L|T)$

+t	+l	0.3
+t	-l	0.7
-t	+l	0.1
-t	-l	0.9

Join T



$P(T,L)$

+t	+l	0.051
+t	-l	0.119
-t	+l	0.083
-t	-l	0.747

Sum out T



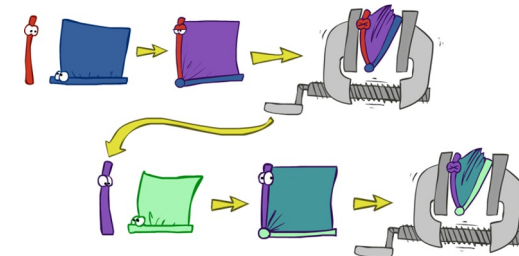
+l	0.134
-l	0.866

General Variable Elimination

- Query: $P(Q|E_1 = e_1, \dots, E_k = e_k)$
- Start with initial factors:
 - Local CPTs (instantiated by *Evidence* if needed)
- While there are still hidden variables (not Q or *Evidence*):
 - Pick a hidden variable H ;
 - Join all factors mentioning H ;
 - Eliminate (i.e., sum out) H .
- Join all remaining factors and normalize.



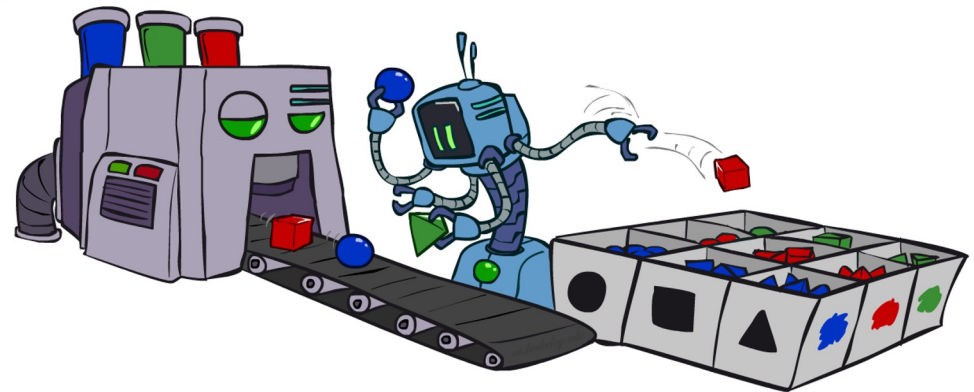
x	$P(x)$
-3	0.05
-1	0.25
0	0.07
1	0.2
5	0.01

$$\text{stick} \times \text{blue square} = \text{purple square} \times \frac{1}{Z}$$

Approximate Inference: Sampling

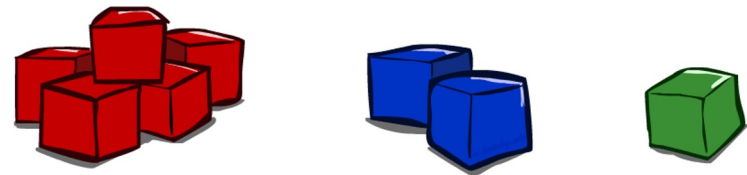
- Variable elimination is for exact inference, which is often impossible.
 - So we need approximate inference, i.e., by sampling.
- Basic idea of sampling for approximate inference:
 - Draw N samples from a sampling distribution S ;
 - Compute an approximate posterior probability;
 - Show this converges to the true probability P .



Sampling from Given Distribution

- Step 1: Get sample u from uniform distribution over $[0, 1)$.
- Step 2: Convert this sample u into an outcome for the given distribution.
 - Associate each target outcome with a sub-interval of $[0,1)$.
 - The sub-interval is size equal to probability of the outcome.

C	$P(C)$
red	0.6
green	0.1
blue	0.3



Example: after sampling 8 times

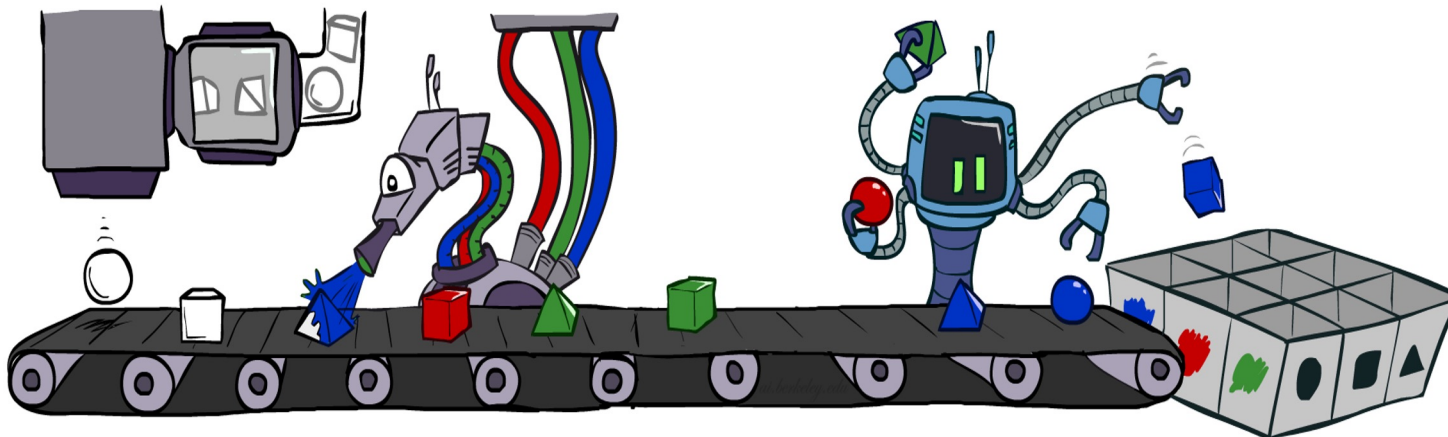
$0 \leq u < 0.6, \rightarrow C = \text{red}$
 $0.6 \leq u < 0.7, \rightarrow C = \text{green}$
 $0.7 \leq u < 1, \rightarrow C = \text{blue}$

Sampling in Bayesian Networks

- Prior Sampling
- Rejection Sampling
- Likelihood Weighting
- Gibbs Sampling

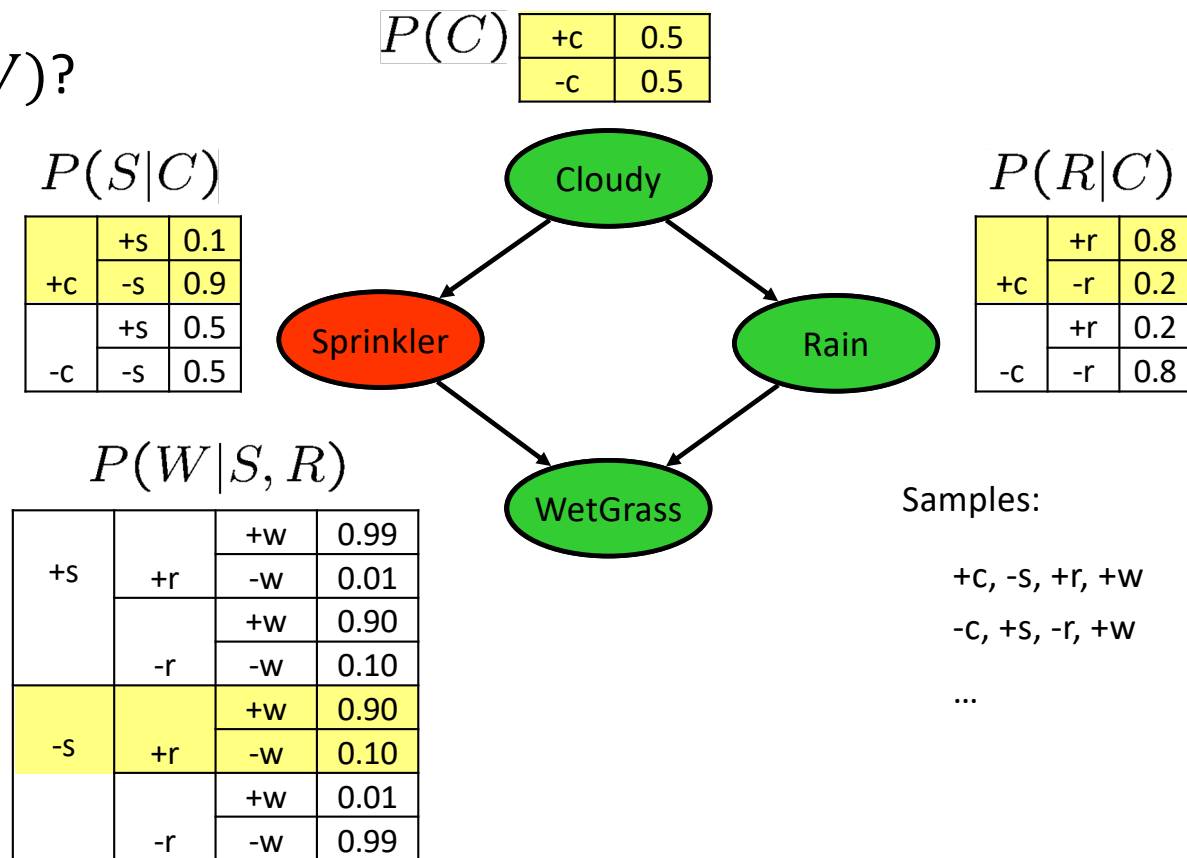
Prior Sampling

For $i = 1, 2, \dots, n$ in topological order
 Sample x_i from $P(X_i \mid \text{Parents}(X_i))$
Return $(x_1, x_2, \dots, x_n)$



Example: Prior Sampling

- What is $P(W)$?



Example: Prior Sampling

- After several sampling, we'll get a bunch of samples from the BN:

+c, -s, +r, +w

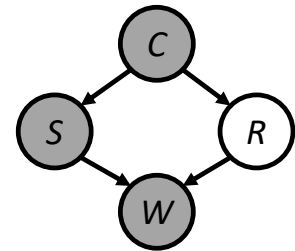
+c, +s, +r, +w

-c, +s, +r, -w

+c, -s, +r, +w

-c, -s, -r, +w

- If we want to know $P(W)$
 - We have counts $\langle +w: 4, -w: 1 \rangle$;
 - Normalize to get $P(W) = \langle +w: 0.8, -w: 0.2 \rangle$;
 - This will get closer to the true distribution with more samples.
- What about $P(C \mid +r, +w)$?
 - Some evidences are instantiated here.



Rejection Sampling

Input: **evidence instantiation**

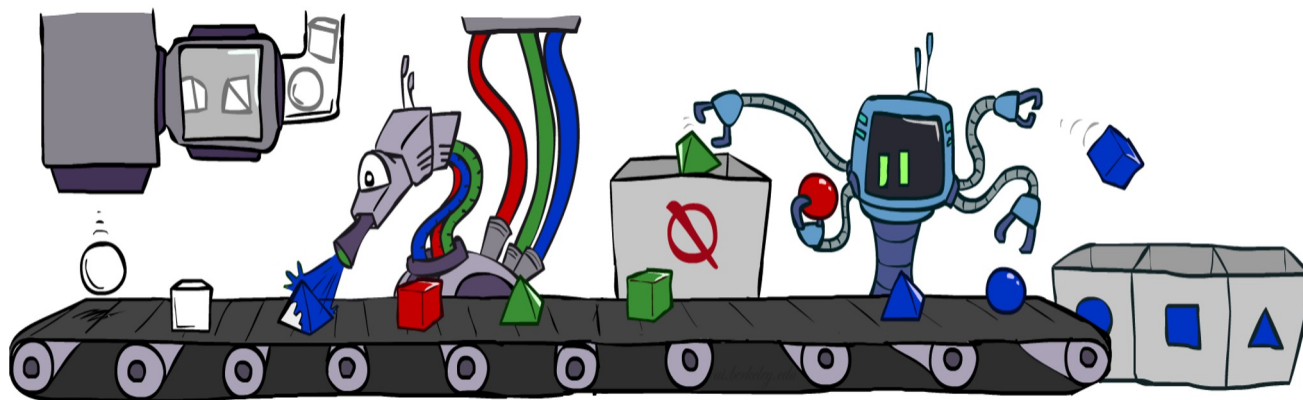
For $i = 1, 2, \dots, n$ in topological order

 Sample x_i from $P(X_i \mid \text{Parents}(X_i))$

 If x_i not consistent with evidence

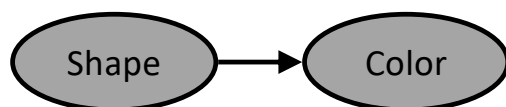
Reject (that means no sample is generated in this iteration)

Return $(x_1, x_2, \dots, x_n)$ (exclude these rejected)



Rejection Sampling

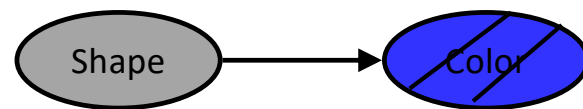
- Problem with rejection sampling:
 - If evidence is unlikely to happen, we will have to reject lots of samples!
 - Example: consider $P(\text{Shape} \mid \text{blue})$



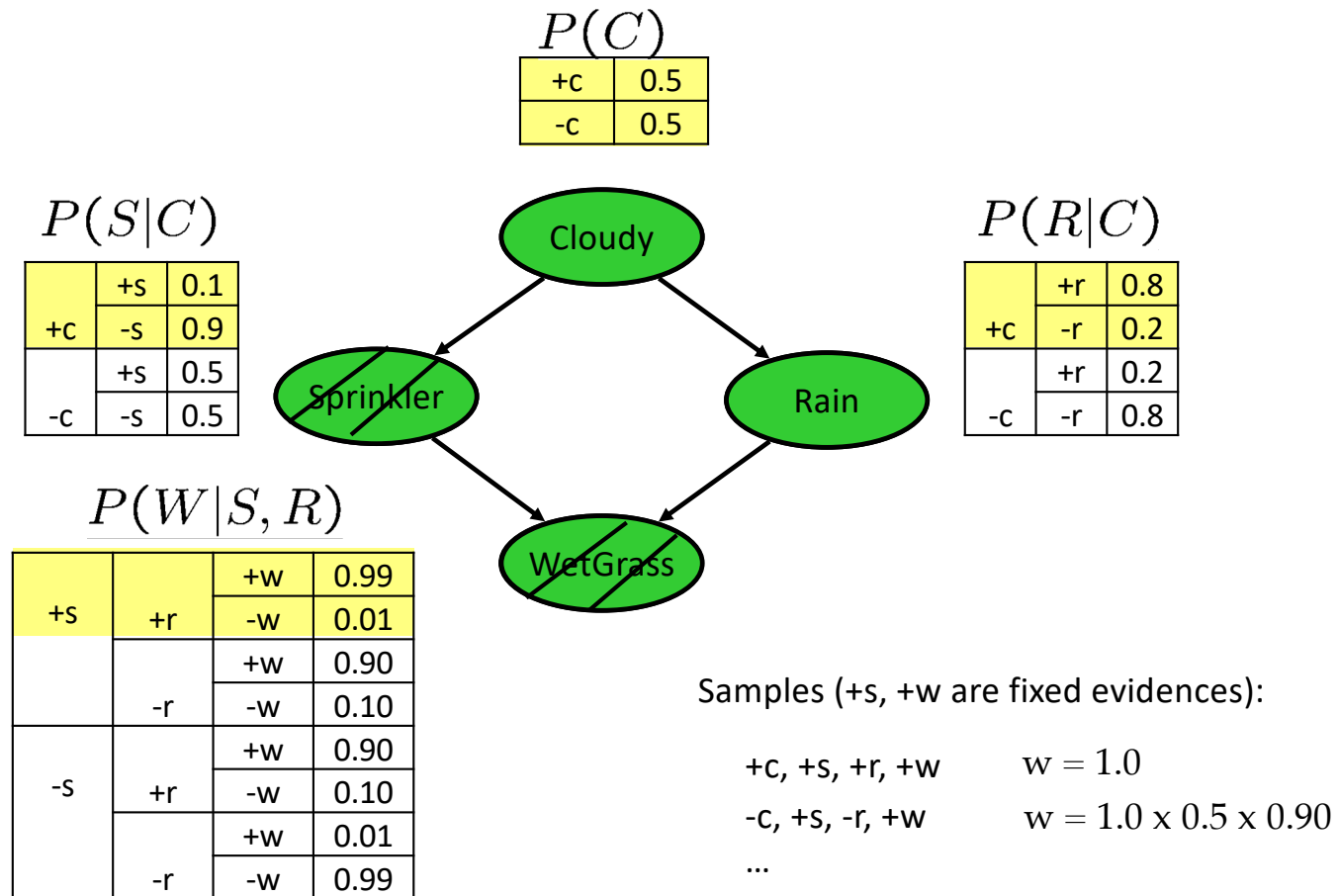
~~pyramid, green~~
~~pyramid, red~~
sphere, blue
~~cube, red~~
~~sphere, green~~

Likelihood Weighting

- Idea: fix evidence variables first, and then sample the rest.
 - Problem: sample distribution is not consistent with real distribution, because of fixed evidence.
 - Solution: weight samples by the probability of the evidence given parents.



pyramid,	blue
pyramid,	blue
sphere,	blue
cube,	blue
sphere,	blue



Likelihood Weighting

Input: evidence instantiation

Initial $w = 1.0$

for $i = 1, 2, \dots, n$ in topological order

 if X_i is an evidence variable

$X_i = \text{observation } x_i \text{ for } X_i$

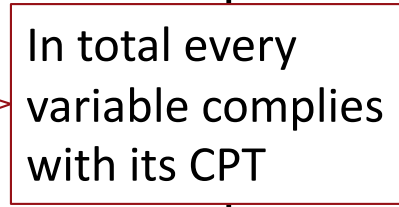
 Set $w = w * P(x_i | Parents(X_i))$

 else

 Sample x_i from $P(X_i | Parents(X_i))$

return $(x_1, x_2, \dots, x_n), w$'s

In total every
variable complies
with its CPT



Likelihood Weighting

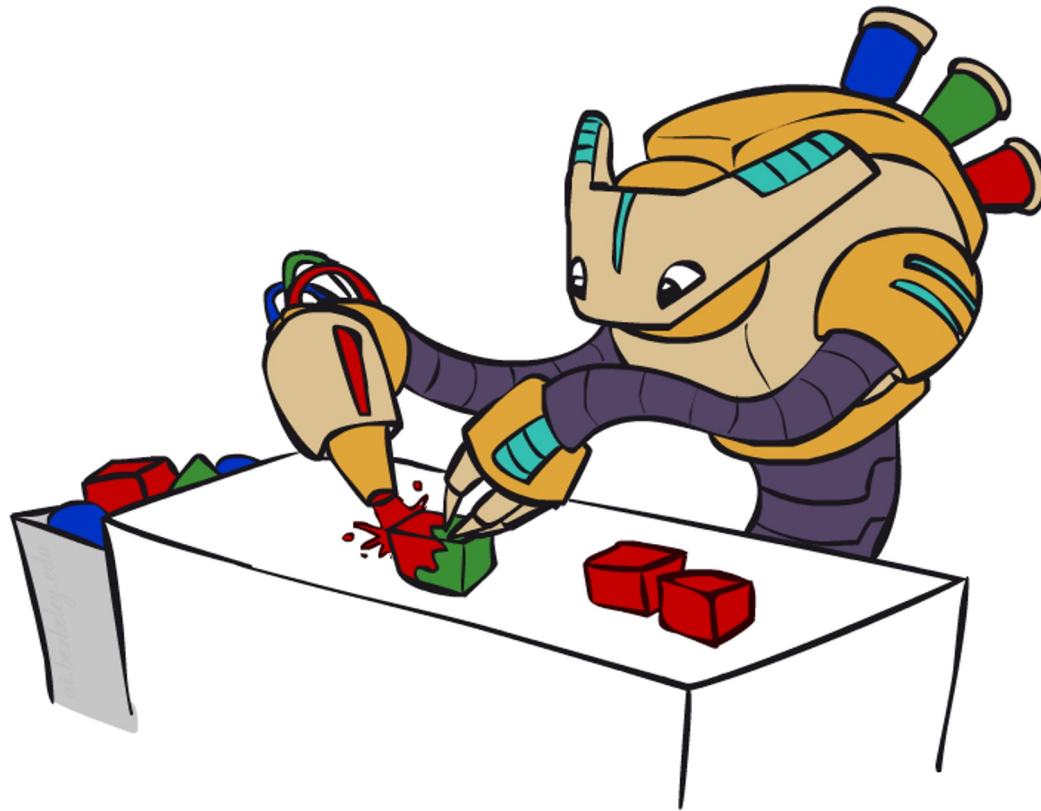
- Pros

- All samples are used.
- More samples will reflect the state of the world suggested by the evidence.
- Values of downstream variables are influenced by upstream evidence.

- Cons

- The values of upstream variables are unaffected by downstream evidence.
- With high probability, one lucky sample will have much larger weight than the others, dominating the result.

Gibbs Sampling



Gibbs Sampling

- We start with a complete assignment of values to all variables in the network.
 - Each variable starts with an initial guess;
 - **Evidence variables** remain fixed throughout the process;
 - **Non-evidence variables** are updated during sampling.

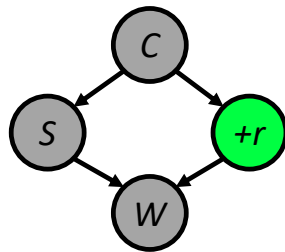
Gibbs Sampling

- Procedure of Gibbs sampling:
 - Initialize all variables with some values;
 - Repeat the following steps:
 - Pick a non-evidence variable X_i ;
 - Sample a new value for X_i from its conditional distribution, given all other current variable values: $X'_i \sim P(X_i | x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$.
 - In a Bayes net, $P(X_i | x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = P(X_i | \text{MarkovBlanket}(X_i))$.
- After enough iterations, the sequence of sampled states will approximate the true joint distribution $P(X)$.
 - These samples can then be used to estimate probabilities, expectations, or make predictions.

Example: Gibbs Sampling

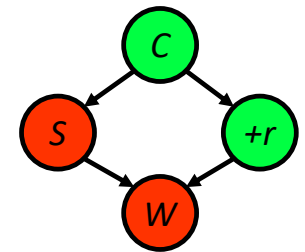
1: Fix evidence

- $R = +r$



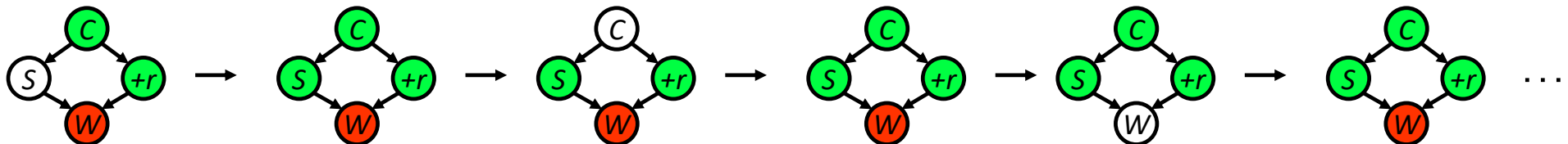
2: Initialize other variables

- Randomize



3: Repeat:

- Choose a non-evidence variable X
- Resample X from $P(X \mid \text{MarkovBlanket}(X))$



Sample from $P(S \mid +c, -w, +r)$

Sample from $P(C \mid +s, -w, +r)$

Sample from $P(W \mid +s, +c, +r)$

Markov Chain Monte Carlo

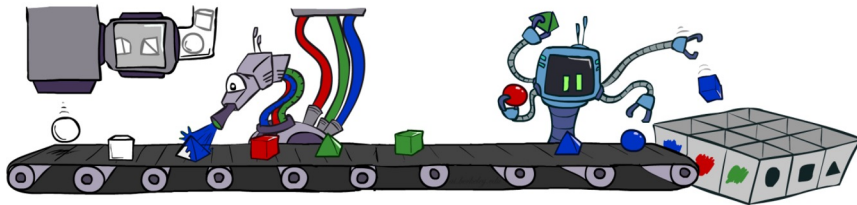
- **MCMC** methods are used to approximate complex probability distributions by generating samples, especially when exact computation is infeasible due to a large state space.
- **Markov Chain**
 - A sequence of states where each state is chosen based only on the previous one.
 - E.g., A full assignment to all variables in Gibbs sampling can be perceived as a state.
- **Monte Carlo**
 - A sampling-based algorithm that uses randomness to estimate answers.
 - May produce slightly inaccurate results, but becomes reliable with enough samples.

Gibbs Sampling: MCMC Example

- Gibbs Sampling is a specific type of MCMC method.
 - Why Monte Carlo?
 - It uses randomness to generate samples;
 - It approximates expectations or probabilities using those samples.
 - It improves sampling by using a Markov chain to generate samples that are dependent but still converge to the target distribution.
 - This avoids the inefficiency of naive random sampling in high-dimensional spaces.

Summary: Bayes Net Sampling

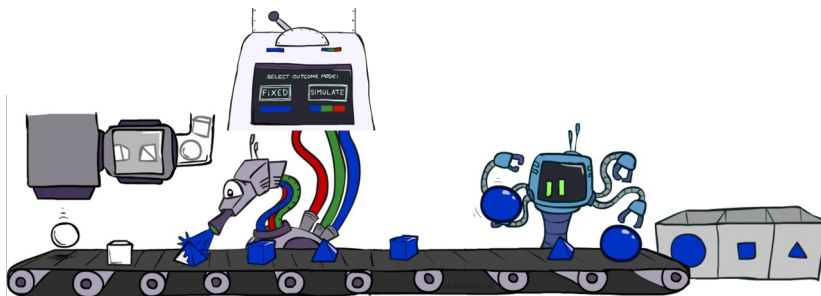
Prior Sampling $P(Q)$



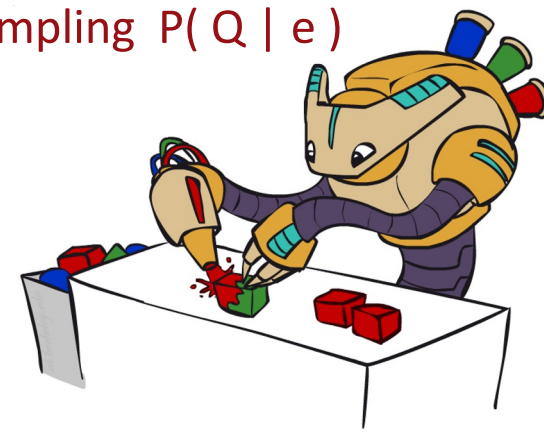
Rejection Sampling $P(Q | e)$



Likelihood Weighting $P(Q | e)$



Gibbs Sampling $P(Q | e)$



Thank You!